HuddleAI

Product Requirements Document

Version 1.6.2 | April 2026

Product	HuddleAI
Version	1.6.2
Status	Active Development
URL	huddleai.app
Stack	React, TypeScript, Firebase, OpenAI GPT-4, Google Cloud Speech-to-Text
Last Updated	April 2026

1. Overview

HuddleAI is an AI-powered meeting intelligence layer. It automatically processes, transcribes, and summarizes meetings — then delivers those insights directly into the tools teams already use.

HuddleAI is NOT a project management tool. It does not compete with Jira, Linear, Notion, or Asana. It integrates with them. The product's value is in making meeting intelligence invisible — action items appear in Jira, summaries land in Notion, and teams barely notice a new tool has been added.

2. Problem Statement

Teams of all sizes lose significant time and context after meetings. Action items get lost, decisions go undocumented, and follow-up is inconsistent. Existing solutions either:

- Require teams to adopt entirely new project management workflows (high friction, low adoption)
- Produce summaries that live in a silo — disconnected from where work actually happens
- Are built exclusively for large enterprise teams, ignoring individual and casual users

The result is that meeting intelligence never makes it into the tools people depend on daily.

3. Product Vision

Make meeting outcomes automatically flow into every tool a team already uses — with zero extra effort from users. And for individuals, make every meeting instantly searchable, summarized, and shareable.

Design Principle

For every feature decision, ask: does this make meeting intelligence flow better into tools teams already use, or does it make the solo experience simpler and more useful? If neither, don't build it.

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4. Target Audience

Solo / Casual Users — Free Forever

- Individuals who want to capture and summarize their own meetings
- Non-technical users — friends, freelancers, students, small business owners
- No team setup required — start immediately after signup
- Gets: transcription, plain text summary, action items list, export. No integrations.
- Upgrades to team features whenever ready — never prompted

Team Users (Free) — Friends & Small Projects

- Solo users who want to share meeting summaries with a few friends or teammates
- Small projects, side projects, early-stage teams with no budget
- Gets: everything in Solo, plus shared team workspace and teammate invites. No integrations.

Team Users (Paid) — Professional Teams

- Tech-forward engineering and product teams using Jira, Linear, Notion, Google Workspace
- Want full automation — meetings processed, action items pushed to Jira, summaries in Notion
- Gets: everything in Team Free, plus all integrations, auto-ingest, admin controls, SSO

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5. Pricing Tiers

Plan	What's Included
Solo	Upload MP3/MP4 (up to 1GB). AI transcription and plain text summary. Action

Free forever	items list. Export/copy summary. English only. No integrations.
Team (Free) Free forever	Everything in Solo. Shared team workspace. Invite teammates — no credit card required. No integrations. Ideal for friends and small projects.
Team (Paid) Paid / per team	Everything in Team Free. Full integration suite: Slack, Jira, Linear, Asana, Notion, Confluence, Google Drive. Auto-ingest from Google Meet and Zoom. Searchable transcript history. Recurring meeting context. Admin controls and SSO.

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6. Out of Scope

The following are explicitly out of scope and will not be built:

- Scrum boards or kanban views
- Internal task assignment and status tracking within HuddleAI
- Sprint planning or backlog management
- Any feature that positions HuddleAI as a standalone project management tool
- Multi-language support (English only for MVP)

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7. Technical Constraints (MVP)

- Supported file formats: MP3 (audio) and MP4 (video)
- Maximum file size: 1GB
- Language support: English only
- Transcription: Google Cloud Speech-to-Text
- AI summarization: OpenAI GPT-4

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8. Onboarding & User Flow

Every new user starts as a Solo user by default. There is no team setup prompt, no onboarding wizard, no friction. The first experience is:

- Sign up → land on dashboard → upload a meeting → get a summary
- No team creation required to get value
- Team features (shared workspace, integrations) are available in settings whenever the user is ready
- Upgrading to a team plan is self-serve — user invites teammates at their own pace

The goal is that a non-technical user (a friend, a freelancer, a student) can sign up and get a useful summary within 5 minutes, without ever thinking about teams or integrations.

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9. Action Items — Reframe

The existing action items feature is valuable but needs to be repositioned. The current implementation has CRUD-style task management built in, which conflicts with the product vision.

What to Keep

- AI extraction of action items from meeting transcripts
- Display of extracted action items in meeting detail view
- Simple action items list across all meetings — not a kanban, just a clean unified view
- Push action items to external tools (Jira, Linear, Asana, Slack) for paid team users

What to Remove or Reframe

- Internal status tracking (mark complete, in progress, etc.) — this is project management
- Internal assignment within HuddleAI — assignees belong in Jira, not HuddleAI
- Any UI that makes action items feel like a task manager

The reframe: action items are meeting outputs, not tasks. The primary CTA on every action item is 'Send to Jira' or 'Send to Linear' — not 'Mark complete.'

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10. Current State (v0)

The following features are already built and deployed at huddleai-a812c.web.app (migrating to huddleai.app):

- Meeting upload (MP4 video files up to 1GB)
- Automatic audio extraction via FFmpeg
- Transcription with speaker diarization (Google Cloud Speech-to-Text)
- AI-generated summaries, action items, topics, decisions (GPT-4)
- Slack integration — manual and automatic meeting notifications
- Team workspaces with role-based access control
- Firebase Auth, Firestore, Cloud Functions infrastructure
- Custom domain: huddleai.app (Cloudflare + Firebase Hosting)

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11. Phased Roadmap

Phase	Focus	Key Deliverables
Phase 1	**Focus & Clean Up**	Remove /team-setup redirect — new users land on /dashboard. Remove internal task management UI. Reframe action items as meeting outputs. Add MP3 support (storage rules + FFmpeg). Solo onboarding flow.
Phase 2	**Auto-Ingest + Email**	Email delivery of summaries (SendGrid/Resend — 1-2 weeks). Google Calendar + Meet OAuth (single auth). Auto-pull recordings. Manual upload remains for all users.
Phase 2.1	**Auto-Ingest Expand**	Zoom integration. Microsoft Teams integration.
Phase 3	**Integration Layer**	Jira v1 (manual trigger, field mapping UI). Linear and Asana (Phase 3.1+). Notion/Confluence summary sync. Google Drive archiving.
Phase 4	**Intelligence Layer**	Full-text search via Algolia or Typesense. Recurring meeting context. Team-level analytics. Weekly digest.

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12. Technical Stack

Frontend	React, TypeScript, Tailwind CSS, shadcn/ui
Backend	Firebase Cloud Functions (Node.js, TypeScript)
Database	Firestore
Storage	Firebase Storage
Auth	Firebase Authentication
AI / ML	OpenAI GPT-4, Google Cloud Speech-to-Text
Hosting	Firebase Hosting + Cloudflare (huddleai.app)
Search (Phase 4)	Algolia or Typesense (Firestore full-text not suitable)
Region	us-central1

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13. Success Metrics

Solo Tier

- Time from signup to first processed meeting (target: under 5 minutes)
- % of solo users who process at least 3 meetings in first 30 days

Team Adoption

- % of team users who connect at least one integration within 30 days (target: >60%)
- % of meetings processed via auto-ingest vs. manual upload (target: >70% post-Phase 2)

Retention

- Weekly active users at 30 / 60 / 90 days
- Teams with 2+ integrations connected (strongest retention signal)

Upgrade

- % of solo users who upgrade to a team plan within 90 days

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14. Engineering Instructions for Phase 1

The following instructions are strict directives for the AI-assisted development environment (Antigravity). Follow them in order. Do not skip steps or reorder them. Each task has a clear definition of done.

Task 1 — Remove /team-setup Redirect [Day 1, P0]

This is the first thing to fix. It is blocking the entire solo user vision.

What to do

- In App.tsx, find the post-signup redirect that routes new users to /team-setup
- Replace it with a redirect to /dashboard
- Ensure /team-setup still exists as a route (accessible from Settings) but is never shown automatically
- Remove any onboarding wizard or modal that appears on first login

Definition of done

- New user signs up → lands on /dashboard with upload CTA
- /team-setup is accessible via Settings but never triggered automatically
- No regression for existing users who already have teams

Task 2 — Remove Internal Task Management UI [P0]

Remove all UI that makes HuddleAI feel like a task manager. Do not remove the data — only the UI.

What to remove

- Scrum board component and its route
- 'Mark Complete', 'In Progress', 'To Do' status controls on action items
- Internal assignee field on action items (within HuddleAI — not the extracted assignee text)
- Any kanban or board view

What NOT to remove

- The action items list view — keep it
- AI-extracted assignee names displayed as text — keep them
- Action items displayed in meeting detail view — keep them

Definition of done

- No status tracking UI exists anywhere in the app
- Action items are displayed as a read-only list with extracted context
- No regression on meeting processing or summary display

Task 3 — Reframe Action Items as Meeting Outputs [P0]

Change the primary CTA on every action item from internal tracking to external push. In Phase 1, these CTAs are placeholders — the actual integrations are built in Phase 3.

What to build

- Each action item gets two placeholder buttons: 'Send to Jira' and 'Send to Linear'
- Clicking either shows a toast: 'Jira integration coming soon — upgrade to Team (Paid)'
- Add 'Copy to clipboard' as a functional action on each action item
- Add 'Copy all action items' button at the top of the action items list

Definition of done

- Every action item has Copy, Send to Jira (placeholder), Send to Linear (placeholder)
- No 'Mark Complete' or status UI anywhere
- Copy to clipboard works correctly

Task 4 — Add MP3 Support [P0]

The current pipeline only handles MP4. Extend it to accept MP3 audio files.

What to change

- Firebase Storage rules: add audio/mpeg to allowed MIME types alongside video/*
- Upload UI: update file input validation to accept .mp3 in addition to .mp4
- Upload UI: update help text to say 'Supports MP3 and MP4, up to 1GB'
- Cloud Function (processMeetingUpload): detect MP3 input and handle accordingly — MP3 does not need audio extraction via FFmpeg, pass directly to Speech-to-Text
- Test: upload a real MP3 file end-to-end and confirm transcript + summary are generated

Definition of done

- MP3 files upload without error
- MP3 files are transcribed and summarized correctly
- MP4 files continue to work without regression
- Files over 1GB are rejected with a clear error message

Task 5 — Solo Onboarding Flow [P0]

The first-time experience for a new user should be: sign up → see dashboard → upload a meeting. Nothing else.

What to build

- Empty state on /dashboard for new users: headline 'Upload your first meeting', subtext 'Get an AI summary in minutes', large Upload button
- Remove any team-related prompts, modals, or banners from the new user flow
- 'Invite teammates' option lives in Settings → Team, not in onboarding
- No changes to the signup form itself — email + password is fine

Definition of done

- New user lands on /dashboard after signup with clear upload CTA
- No team setup prompt appears at any point in the default flow
- Existing users with teams are not affected

General Rules for All Phase 1 Work

Do not build

- Email delivery — this is Phase 2. Do not add any email infrastructure in Phase 1.
- Any new integrations — Jira, Linear, Slack changes are Phase 3. Placeholders only.
- Auto-ingest — this is Phase 2. Manual upload remains the only input method in Phase 1.

Code standards

- All changes in TypeScript — no plain JS
- No new Firestore collections without explicit approval
- Do not modify the Cloud Function processing pipeline beyond MP3 support

- All UI changes must be mobile-responsive
- Deploy to huddleai.app after each task is complete and tested

Testing requirements

- Each task must be manually tested before marking done
- Test both solo user flow AND existing team user flow for every change
- No regressions on meeting upload, processing, or summary display are acceptable

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15. User Flow — Solo to Team

Every user starts as Solo regardless of intent. The product earns trust through the core value loop before asking for anything. Team features are discovered, not forced.

Stage 1 — Solo Empty (zero meetings, no team)

This is every new user's first experience. The entire screen is focused on one action.

- Dashboard shows: large headline 'Upload your first meeting', subtext 'Get an AI summary in minutes', single Upload button
- No stats, no team UI, no Slack prompt, no integration banners
- Header: HuddleAI logo + Upload Meeting button only
- No mention of teams, plans, or integrations anywhere on screen

Stage 2 — Solo Active (has meetings, no team)

User has uploaded at least one meeting. They've seen the core value. Dashboard becomes useful.

- Dashboard shows: recent meetings list, basic stats (total meetings, total duration)
- Action items list accessible from nav — clean read-only list, no task management
- No team UI, no Slack prompt, no integration banners
- Subtle 'Invite a teammate' link in Settings only — never surfaced on the main dashboard
- Header: logo + Upload Meeting button

Stage 3 — Team Free (has team, free plan)

User has invited at least one teammate. Shared workspace is now relevant.

- Dashboard label changes to 'Team Dashboard'

- Shared meetings visible to all team members
- Team Members section visible in dashboard
- Slack Integration prompt now surfaces — it's relevant at this stage
- Jira/Linear CTAs on action items show 'Coming soon — upgrade to Team (Paid)'
- Stats reflect the whole team's activity

Stage 4 — Team Paid (has team, paid plan)

User has upgraded. Integrations are unlocked. This is the full product experience.

- All integrations active: Slack, Jira, Linear, Asana, Notion, Confluence, Google Drive
- 'Send to Jira' and 'Send to Linear' CTAs are functional, not placeholders
- Auto-ingest settings visible (Phase 2+)
- Admin controls visible to team owner
- Full stats and analytics dashboard

userStage Utility — Engineering Spec

The dashboard must read a userStage value and render the correct view. Build a userStage utility function that returns one of four string constants based on the following logic:

- SOLO_EMPTY — user has 0 meetings AND is not a member of any team
- SOLO_ACTIVE — user has 1+ meetings AND is not a member of any team
- TEAM_FREE — user is a member of a team AND team plan is free
- TEAM_PAID — user is a member of a team AND team plan is paid

The dashboard component imports userStage and renders the appropriate view. No other component should contain stage-branching logic — it all flows from this single utility.

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16. Engineering Instructions for Phase 2

Sprint 2A is complete. Phase 2 continues with Google Meet + Calendar auto-ingest as the next priority. Email delivery has been formally removed from the roadmap — see Section 17 for the decision record.

Sprint 2A — Dashboard userStage + UI Rework [✅ COMPLETE]

Sprint 2A is fully built and deployed. The following was delivered:

- userStage utility built in src/lib/userStage.ts — single source of truth for user state

- resolveUserStage(userId, team) returns {stage, soloMeetings} — handles all 4 stages
- SOLO_EMPTY → centered 'Upload your first meeting' CTA
- SOLO_ACTIVE → meetings list with status badges + subtle 'Create Team' nudge card
- TEAM_FREE / TEAM_PAID → full existing dashboard
- teamId made nullable throughout — solo users can upload without a team
- Storage path falls back to userId when no team: meetings/{userId}/{meetingId}.ext
- Routing fixed: all auth methods (email, Google login, Google signup) always → /team, never /team-setup

Sprint 2B — Google Meet + Calendar Auto-Ingest [P0, Next]

This is the highest priority remaining feature. It eliminates the biggest friction point in the product — manual file upload. The goal is for Google Meet users to never have to think about uploading a recording again.

How it works — Processing Strategy

There are two ways to get meeting content from Google. Use them in this priority order:

- Strategy 1 — Transcript first: Google Meet natively generates transcripts on Google Workspace Business Standard and above. If a transcript is available, pull it directly and pass to GPT-4 for summarization. This skips Speech-to-Text entirely — faster, cheaper, no audio processing overhead.
- Strategy 2 — Recording fallback: If no transcript is available (personal Gmail accounts, Business Starter plan), fall back to pulling the video recording from Google Drive and running it through the existing FFmpeg + Speech-to-Text pipeline.
- Strategy 3 — Manual upload fallback: Always available for all users regardless of Google connection. Never remove this path.

This tiered approach means: paid Google Workspace users (most likely to be HuddleAI Team Paid users) get the fastest, cheapest path. Personal Gmail users get the recording-based path. Everyone can always fall back to manual upload.

Google Workspace Transcript Availability — Important Context

- Transcripts available: Business Standard ($12/user/mo), Business Plus ($18/user/mo), Enterprise
- Transcripts NOT available: Free personal Gmail (@gmail.com), Business Starter (cheapest paid tier)
- In practice: company email users on Workspace → transcript path. Personal Gmail users → recording path.
- This maps cleanly to HuddleAI tiers: Team Paid users are likely on Workspace, Solo users are likely on personal Gmail

What to build

- Google OAuth 2.0 flow — scopes required: calendar.readonly, drive.readonly, drive.metadata.readonly. If transcript access is available add: https://www.googleapis.com/auth/meetings.space.readonly
- 'Connect Google Account' button in Settings → Integrations (available to TEAM_FREE and TEAM_PAID only — not solo users)
- Store OAuth tokens securely in Firestore — encrypted, per-user, never in localStorage or client-side
- Token refresh handler — OAuth tokens expire, implement automatic refresh so users never need to reconnect
- After OAuth: listen for completed meetings via Google Calendar API push notifications (preferred over polling — more efficient)
- On meeting end: check for transcript first via Google Meet API or Drive. If found → extract text → pass to GPT-4 summary pipeline
- If no transcript: check for recording in Google Drive → download → pass to existing FFmpeg + Speech-to-Text pipeline
- Source label on each meeting in dashboard: 'Google Meet (Transcript)', 'Google Meet (Recording)', or 'Uploaded' so users understand how it was processed
- Disconnect button in Settings — revokes OAuth tokens and stops auto-ingest cleanly

What NOT to build in this sprint

- Zoom integration — Phase 2.1, after Google Meet is stable and tested
- Microsoft Teams integration — Phase 2.2
- Email delivery — formally removed from roadmap (see Section 17)
- Any changes to the Jira/Linear integration placeholders — Phase 3
- Any changes to the AI summarization pipeline beyond accepting transcript text as input

Definition of done

- User connects Google account once in Settings → all future Google Meet recordings are automatically processed
- Transcript path works end-to-end for Google Workspace users — meeting ends → summary appears in dashboard within 5 minutes
- Recording fallback path works end-to-end for personal Gmail users — meeting ends → recording pulled → processed within 10 minutes
- Each processed meeting has the correct source label in the dashboard
- Manual upload continues to work without regression for all users
- OAuth token refresh is automatic — user never needs to reconnect unless they explicitly disconnect
- Revoking Google access in Settings cleanly stops all auto-ingest and removes stored tokens
- Processing failures from auto-ingest are surfaced in the dashboard — user can see if something failed and trigger manual reprocess

General Rules for Phase 2 Remaining Work

Code standards

- All OAuth tokens stored encrypted in Firestore — never in localStorage or client-side storage
- All Cloud Function changes must handle errors gracefully — auto-ingest failures must never break manual upload or existing processing
- All new UI must respect userStage — no hardcoded team/solo assumptions in any component
- All changes in TypeScript — no plain JS
- Mobile-responsive for all new UI
- Test locally first — only deploy when the sprint is fully complete and locally validated

Testing requirements

- Test transcript path with a real Google Workspace account before deploying
- Test recording fallback path with a personal Gmail account before deploying
- Test OAuth token expiry and refresh before deploying
- Full regression test on manual upload flow after deploy
- Test all four userStage dashboard views still render correctly after any dashboard changes

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17. Decision Record

This section documents significant product decisions and the reasoning behind them. These are final decisions — they should not be revisited without a strong new reason.

Decision: Email Delivery Removed from Roadmap

What was decided

Email delivery of meeting summaries has been permanently removed from the roadmap. It will not be built unless users explicitly request it at scale.

Why

- HuddleAI already has better delivery channels: in-app (immediate, right where the user processed the meeting) and Slack (delivered where team conversations happen)
- Email is a passive, low-engagement channel. Team users won't act on action items from an email — they'll act on them in Jira or Slack.
- Solo users who just uploaded a meeting are already in the app — they'll see the summary immediately without needing an email

- Email deliverability requires SPF/DKIM DNS configuration, third-party API setup, and cross-client HTML template testing — significant engineering cost for low product value
- Engineering time is better spent on Google Meet auto-ingest, which directly removes friction from the core user flow

What replaces it

- In-app: summary is displayed immediately when processing completes
- Slack: team users get rich notifications with summary and action items in their channel
- Future: push notifications (mobile) can be considered if HuddleAI ships a mobile app

Decision: Transcript-First Processing for Google Meet

What was decided

Google Meet auto-ingest will attempt to use Google's native transcript first, falling back to recording-based processing only when a transcript is unavailable.

Why

- Google Meet transcripts are already high quality — using them skips the expensive Speech-to-Text processing step entirely
- Transcript-based processing is faster (no audio download, no FFmpeg, no Speech-to-Text latency) and cheaper (no Speech-to-Text API costs per minute)
- The users most likely to have transcripts available (Google Workspace Business Standard+) are the same users most likely to be on HuddleAI's paid team plan — the highest-value user segment
- Recording fallback ensures the feature still works for personal Gmail users and lower-tier Workspace plans

Decision: No Standalone Kanban or Task Management

Documented from v1.0. HuddleAI does not build internal task management. Action items are meeting outputs to be pushed to external tools — not tracked internally. The primary CTA on every action item is 'Send to Jira' or 'Send to Linear', not 'Mark complete.'